Practice workbook · Quadratic inequalities

Rearranging an inequality into the form ax^2 + bx + c < 0 (or > 0), finding the critical values and choosing the correct region.

10 question parts: 1 worked-example retry and 9 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-IDN-02

1. Question 1 · part b · 4 marks

2023 JAN · Paper 02

Worked-example retry · Attempt part b only.

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1 Find the set of values of x for which

(a) $2(3x - 1) < 4 - 3x$

(2)

(b) $3x^2 - 8x - 3 < 0$

(4)

(c) **both** $2(3x - 1) < 4 - 3x$ **and** $3x^2 - 8x - 3 < 0$

(1)

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(Total for Question 1 is 7 marks)

1. Question 1 · part b · 4 marks · Working and answer

Working and answer

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2. Question 2 · part b · 4 marks

2022 JAN · Paper 01

Further practice · Attempt part b only.

2 $f(x) = 2x^2 - 12x + 5$

Given that $f(x)$ can be written in the form $f(x) = a(x + b)^2 + c$ where a, b and c are integers,

(a) find the value of a, the value of b and the value of c

(3)

(b) Hence find the set of values of x for which $f(x) - 37 > 0$

(4)

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P 6 6 3 1 0 A 0 4 3 6

2. Question 2 · part b · 4 marks

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Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 7 marks)

2. Question 2 · part b · 4 marks · Working and answer

Working and answer

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3. Question 1 · part b · 3 marks

2022 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 Find the set of values for x for which

(a) $5x - 10 > 4x - 7$

(1)

(b) $2x^2 - 11x + 5 < 0$

(3)

(c) **both** $5x - 10 > 4x - 7$ **and** $2x^2 - 11x + 5 < 0$

(1)

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(Total for Question 1 is 5 marks)

3. Question 1 · part b · 3 marks · Working and answer

Working and answer

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4. Question 2 · part b · 4 marks

2023 JAN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $3x - 8 < 5x + 3$

(1)

(b) $4x^2 - 7x + 1 > 6 - 2x^2$

(4)

(c) **both** $3x - 8 < 5x + 3$ **and** $4x^2 - 7x + 1 > 6 - 2x^2$

(1)

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4. Question 2 · part b · 4 marks · Working and answer

Working and answer

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5. Question 4 · part b · 4 marks

2024 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4

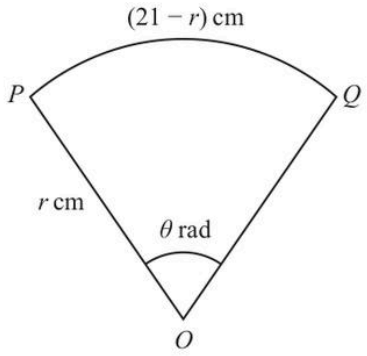


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows the sector OPQ of a circle with centre O and radius r cm.

$OP = OQ = r$ cm arc $PQ = (21 - r)$ cm $\angle POQ = \theta$ radians

The area of the sector is A cm^2

(a) Show that $A = \frac{r}{2}(21 - r)$ **(3)**

The area of the sector must be greater than or equal to 27 cm^2

(b) Find the set of possible values of r **(4)**

(c) Hence write down the set of possible values of θ **(2)**

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5. Question 4 · part b · 4 marks · Working and answer

Working and answer

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6. Question 2 · part b · 3 marks

2025 JUN · Paper 02

Further practice · Attempt part b only.

2 Find the set of values for x for which

(a) $9-3x>11x+2$

(1)

(b) $10x^2+7x<12$

(3)

(c) **both** $9-3x>11x+2$ **and** $10x^2+7x<12$

(1)

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6. Question 2 · part b · 3 marks · Working and answer

Working and answer

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7. Question 1 · part b · 3 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

1 Find the set of values for x for which

(a) $8x - 7 < 5x + 5$

(2)

(b) $2x^2 - 5x - 3 > 0$

(3)

(c) **both** $8x - 7 < 5x + 5$ **and** $2x^2 - 5x - 3 > 0$

(1)

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(Total for Question 1 is 6 marks)

7. Question 1 · part b · 3 marks · Working and answer

Working and answer

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8. Question 2 · part b · 3 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $2(x+1) < 5x - 2$

(2)

(b) $3x^2 - x \leqslant 10$

(3)

(c) **both** $2(x+1) < 5x - 2$ **and** $3x^2 - x \leqslant 10$

(1)

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8. Question 2 · part b · 3 marks · Working and answer

Working and answer

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9. Question 2 · part b · 4 marks

2020 JAN · Paper 02

Further practice · Attempt part b only.

2 Find the set of values of x for which

(a) $3 + 2x \leqslant x + 2$

(1)

(b) $8x^2 + 10x < 3$

(4)

(c) **both** $3 + 2x \leqslant x + 2$ **and** $8x^2 + 10x < 3$

(1)

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P 5 9 9 3 9 A 0 4 3 6

9. Question 2 · part b · 4 marks · Working and answer

Working and answer

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10. Question 2 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

2 (a) Use algebra to find the x coordinates of the points where the curve with equation $y = 3x^2 + 9x - 17$ intersects the line with equation $y = 3 - 2x$

(3)

(b) Hence, or otherwise, find the set of values of x for which $3 - 2x \leqslant 3x^2 + 9x - 17$

(2)

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P 7 8 9 6 5 A 0 4 3 2

10. Question 2 · part b · 2 marks

2025 NOV · Paper 02

Further practice · Attempt part b only.

Question 2 continued

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(Total for Question 2 is 5 marks)

10. Question 2 · part b · 2 marks · Working and answer

Working and answer

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